Technical Document Nurse Call System

Introduction

The N-Bed Nurse Call System is designed to provide communication between patients and nurses in a hospital environment. Each bed is equipped with a patient unit containing four buttons: Nurse Call, Medicine Request, Acknowledge, and Cancel. The nurse station consists of a dashboard that displays the bed number, patient name, request type, timestamp, and status of all active requests.

System Architecture

The N-Bed Nurse Call System follows a distributed client-server architecture. Each patient bed is equipped with an ESP32-based patient unit, while a central ESP32 acts as the nurse station and web server.

The patient units communicate with the nurse station through a common Wi-Fi network using HTTP requests. Each patient unit is identified using a unique room/bed number and patient information.

The nurse station receives requests from multiple patient units, maintains the state of each request, records timestamps, calculates response time, stores activity logs in RAM, and displays the requests on the nurse station dashboard.

The architecture can be represented as:

```
Patient Unit 1 ──┐
Patient Unit 2 ──┤
Patient Unit 3 ──┤ ── Wi-Fi + HTTP ──> Nurse Station ESP32
Patient Unit N ──┘                          |
                                            ├── Nurse Station Dashboard
                                            └── Administration Dashboard
```

Working of the System

The working of the nurse call system starts in a hospital environment with multiple patients in their respective beds who may require assistance from nurses and hospital staff. Whenever a patient from any bed or room uses the nurse call system by pressing either the Nurse Call button or the Medicine Request button, the nurse station is notified of the request through its display interface.

This interface consists of the bed number, patient name, request type, the time at which the call was initiated, as well as the time at which the particular request was acknowledged by the nurse. This helps the hospital authorities and concerned personnel determine the time taken for a nurse to attend to a patient in need.The system also consists of LEDs, buzzers, and other components to provide both audio and visual alerts for incoming requests. Nurses can acknowledge a request once it has been received and cancel it after the required assistance has been provided.The system is designed to support multiple patient units connected to a single nurse station. Each patient unit is assigned a unique bed or room number, allowing the nurse station to identify and manage requests from different patients independently. When requests from multiple beds are received at the same time, each request is displayed as a separate block on the nurse station interface. The blocks maintain their individual request type, status, patient information, call time, and acknowledgement time. Acknowledging or cancelling one request only affects the corresponding patient block and does not affect other active requests.

Technical Details

The N-bed nurse call system consists of multiple patient units connected to a nurse station through a common Wi-Fi network. Each patient unit contains an ESP32 microcontroller, four touch buttons (Nurse Call, Medicine Request, Acknowledge, and Cancel), an RGB LED, and a buzzer.

The nurse station dashboard is hosted directly on the ESP32 using the WebServer library. The dashboard can be accessed from any laptop, tablet, or smartphone connected to the same Wi-Fi network.

-Patient Unit

Each patient unit consists of an ESP32 microcontroller connected to capacitive touch inputs, an RGB LED, and a buzzer.

The patient unit detects button presses using the ESP32 `touchRead()` function. Each unit is assigned a unique room number and patient name in the firmware.

When a request is generated, the patient unit communicates with the nurse station over the common Wi-Fi network using HTTP requests.

– Nurse Station

The nurse station uses an ESP32 as the central controller and web server.

The nurse station is responsible for:

- Receiving requests from multiple patient units.

- Maintaining the state of active requests.

- Displaying multiple patient request blocks.

- Recording call and acknowledgement timestamps.

- Calculating response time.

- Maintaining active and completed call statistics.

- Storing activity logs in RAM.

- Hosting the nurse station dashboard.

- Hosting the hospital administration dashboard.

Tech Stack

Hardware:

- ESP32
- Touch sensors
- RGB LED
- Buzzer

Software:

- Arduino IDE
- C++
- HTML
- CSS
- ESP32 WebServer library
- Wi-Fi
- NTP server for time synchronization

Network Requirements

All patient units and the nurse station are connected to the same Wi-Fi network.

The nurse station ESP32 acts as the central server, while the patient units communicate with it over the network using HTTP requests.

Each patient unit uses the IP address of the nurse station to send requests. The room number included in each request allows the nurse station to identify which patient unit generated the request.

The current prototype uses a local Wi-Fi network for communication between the ESP32 devices.

HTTP Communication

The patient units communicate with the nurse station using HTTP requests over the local Wi-Fi network.

When a patient generates a request, the corresponding patient unit sends the room number, patient information, and request type to the nurse station.

The nurse station processes the received request and updates the corresponding patient block on the dashboard.

Acknowledgement and cancellation requests are also sent from the respective patient unit to the nurse station. The room number is used to ensure that the correct patient's request is updated without affecting other active requests.

The system currently uses HTTP for communication and does not use JSON for the request data.

Information Flow

1. The patient presses the Nurse Call or Medicine Request button.
2. The ESP32 detects the button press using touchRead()
3. The patient ESP32 sends the room number, patient information, and request type to the nurse station through an HTTP request.
4. The nurse station receives the request, records the call timestamp, and stores the request information.
5. The dashboard displays the request as an independent patient block.
6. The nurse travels to the patient's room and presses the Acknowledge button.
7. The ESP32 records the acknowledgment time and calculates the response time.
8. The response details are stored in the activity log.
9. After assisting the patient, the nurse presses the Cancel button.
10. The request is cleared from the dashboard and the system returns to the idle state.

- Communication protocol: HTTP
- Network: Wi-Fi
- Server: ESP32 WebServer library
- Data format: ESP32 variables (string, int , struct)
- Frontend: HTML and CSS
- The current prototype does not use JSON. Future versions can use JSON payloads for communication between patient units and the nurse station.

Request State Management

Each patient request follows a defined sequence:

IDLE → PENDING → ACKNOWLEDGED → COMPLETED

When a patient initiates a request, its status changes to Pending.

When the nurse acknowledges the request, its status changes to Acknowledged and the acknowledgement time is recorded.

After the required assistance has been provided, the nurse presses Cancel. The request is completed and removed from the active dashboard.

Each request is independently associated with its room number, patient information, request type, status, call time, acknowledgement time, and response time.

Alert and Status Indication

The system provides visual and audio feedback for different request states.

System State	RGB LED / Dashboard Color
Idle	White
Nurse Call	Cyan
Acknowledged	Yellow
Medicine Request	Blue

The buzzer provides an audio alert when a new request is received.

The dashboard uses colored request blocks so that nurses can quickly identify the type and status of requests from multiple patient beds

Response Time Calculation

The system measures the time taken by the nurse to acknowledge a patient request.

When a request is received, the system records the call timestamp.

When the nurse acknowledges the request, the acknowledgement timestamp is recorded.

The response time is calculated as:

Response Time = Acknowledgement Time − Call Time

The response time is displayed in seconds for shorter durations and in minutes and seconds for longer durations.

The calculated response time is also stored in the administration activity log.

Data Storage

The prototype stores active request information and activity logs in the RAM of the nurse station ESP32.

Each activity log contains the room number, request type, call time, acknowledgement time, and response time.

The activity logs are implemented using a C++ structure and an array.

The current prototype does not use a permanent database. Therefore, the stored activity data is lost when the nurse station ESP32 is restarted.

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Backend Block Diagram

Bed 1
ESP32
Nurse Call Button
Medicine Button
Acknowledge Button
Cancel Button
LED, Buzzer

Bed 2
ESP32
Nurse Call Button
Medicine Button
Acknowledge Button
Cancel Button
LED, Buzzer

—--->

Nurse Station
Bed Number
Patient Name
Request Type
Timestamp
Status

.

.

.

N bed units
ESP32
Nurse Call Button
Medicine Button
Acknowledge Button
Cancel Button
LED, Buzzer

Patient Unit
Nurse Call Button
Medicine Button
Acknowledge Button
Cancel Button
RGB LED + Buzzer

↓touch input

ESP32
Detects request
Records timestamps
Calculates response
Maintains logs

↓HTTP over wifi

Nurse Dashboard
Room Number
Patient Name
Request Type
Call Time
Ack Time
Response Time
System Status

↓stores logs in RAM

Temporary Log Memory
Room
Request Type
Call Time
Ack Time
Response Time

Frontend Mock HTML Design

Hospital Nurse Station

Room No	Patient Name	Status	Request Type	Call Time	Acknowledge time
101	John	Idle	-	-	-
102	Jake	Pending	Nurse Call	09:53:26	10:00:32
103	Charlie	Pending	Medicine Requested	09:54:12	-

Administration Dashboard

The administration dashboard provides an overview of the nurse call system activity.

It displays the total number of calls, active calls, completed calls, medicine requests, and recent request activity.

The activity log records the room number, request type, call time, acknowledgement time, and response time.

The dashboard allows hospital administration to monitor request activity and evaluate the response time of nursing staff.

Multi-Bed Operation

The system allows multiple patient units to communicate with a single nurse station simultaneously.

For example, Bed 101 can generate a Nurse Call while Bed 102 generates another request. Both requests are displayed as separate blocks on the nurse station dashboard.

Each patient block maintains its own room number, patient information, request type, status, call time, and acknowledgement time.

Acknowledging or cancelling one request only affects the corresponding patient block and does not affect other active requests.

This allows the system to support an N-bed hospital environment using multiple ESP32 patient units connected to a single nurse station.

NURSE CALL SYSTEM

Room Number : 101

Patient Name : John Doe

Request Type : Nurse Call

Status : Pending

Call Time : 10:15:30 AM

Ack Time : --:--:--

HOSPITAL ADMINISTRATION DASHBOARD

Total Calls Today : 10

Active Calls : 2

Completed Calls : 8

Medicine Requests : 3

Room	Request Type	Call Time	Ack Time	Response
101	Nurse Call	10:15 AM	10:17 AM	2min
102	Medicine	10:20 AM	10:23 AM	3min

Happy Path or Critical Path:

-The patient units and nurse station are powered on and connected to the same Wi-Fi network.

-A patient presses the Nurse Call or Medicine Request button.

-The ESP32 assigned to that bed detects the button press.

-The request is transmitted from the patient ESP32 to the nurse station ESP32 using HTTP over Wi-Fi.

-The dashboard displays the patient's bed number, patient name, request type, and call initiation time.

-The LED and buzzer are activated to alert the nursing staff.

-The nurse views the request on the dashboard and travels to the patient's room.

-Upon reaching the room, the nurse presses the Acknowledge button on the patient unit.

-The acknowledgement signal is transmitted from the corresponding patient ESP32 to the nurse station ESP32 using HTTP over Wi-Fi.

-The dashboard updates the request status to Acknowledged and records the acknowledgment time.

-The nurse provides the required assistance to the patient.

-After completing the assistance, the nurse presses the Cancel button on the patient unit.

-The dashboard removes the request and updates the status to Completed.

-The system returns to the idle state and waits for the next request.